

MAKR 100: Introduction to Microcontrollers and Physical Computing

Project 1

Due Date: September 29, 2026 at 5:00 PM

Read the requirements for each project thoroughly and **select one** to complete for Project 1. If you have questions, contact me. For the project, you will design the circuit, build the circuit on your breadboard, and write the code. We will have one in class workday on September 22, 2026.

Option A: Binary Counter

Using your Arduino/Elegoo UNO, 5 LEDs, a push button, you will create a binary counter display. The counter should count from 0 - 31 in binary using an off LED to denote a 0 and an illuminated LED to denote a 1. Your counter should start at 0 with no LEDs on. Each button press will increment the counter up 1 and illuminate the value with the LEDs. The counter should count to 31 in binary. At that point, pressing the button again will reset the counter back to 0.

If you are unfamiliar with binary numbers, please check out the table [here](#) and review this [article](#).

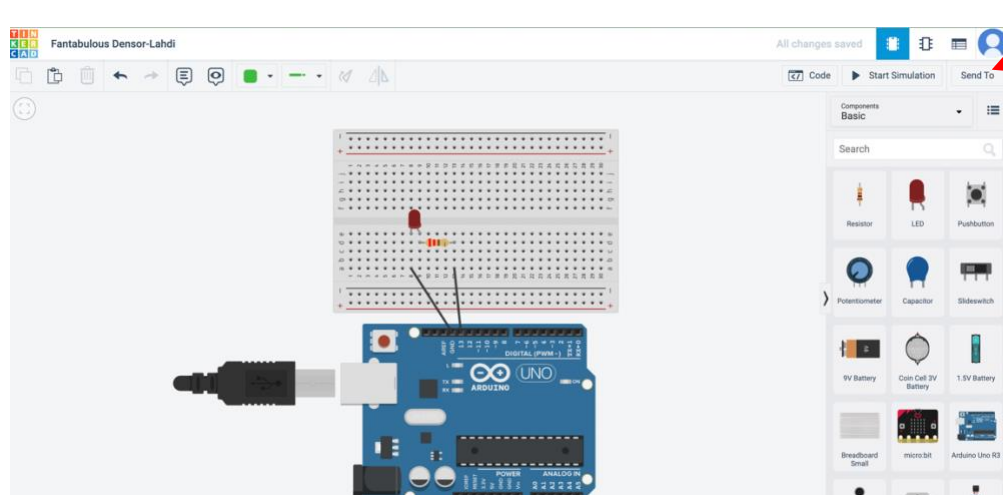
Option B: Logical Operators

Create a project that shows how different logical operators work in real time. This project will utilize two push buttons to represent A and B. Each button will have an indicator LED that will turn on when the button is pressed. An additional four different LEDs will show the output for the following different logical operations: $\neg A$, $A \&\& B$, $A \parallel B$, $(A \&\& \neg B) \parallel (\neg A \&\& B)$. For example, when Button A and Button B are both pressed the indicator LEDs for A, B, and $A \&\& B$ will be on.

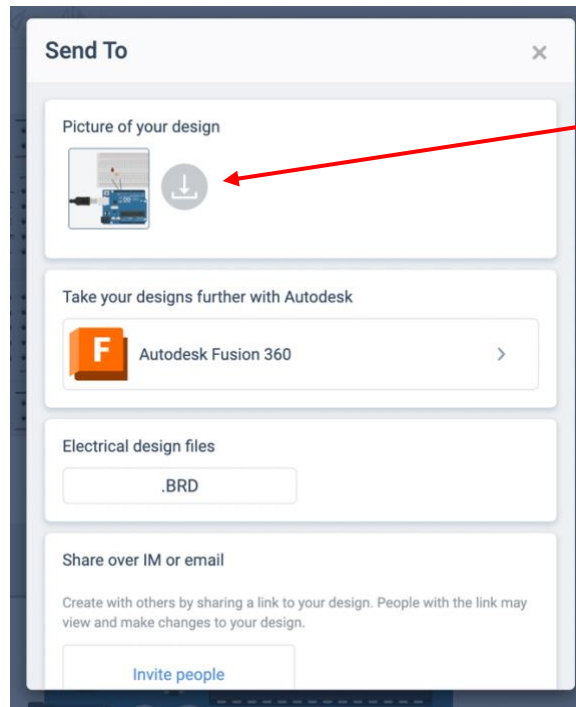
Submission Instructions

Please submit a [zip file](#) title "YourLastName_Project_1.zip" with the following:

1. **Project Report** – this will include a bill of materials (list of parts you used to build your circuit), description of your solution, and any challenges or interesting takeaways. There is no word count limit, but I would expect that you use as many words as necessary so that someone else could reproduce your project. Do not forget to include your name.
2. **Project Schematic** – Use Tinkercad to show me how your circuit is connected. In Tinkercad, you can download an image of your design. To do this, click the "Send To" button in the upper-right hand corner of the Tinkercad menu (pictured below to the left of the red arrow).



Click the download button on the “Send To” screen (indicated by the red arrow).



3. **Video Demo** – A video showing your project in action. Show that the various requirements of the project are fulfilled. Either include in the video in the zip file or upload to YouTube and include the link in your report. Make sure your YouTube video(s) is/are public for this project.
4. **Project Code/Sketch** – **.ino** source code file. The code should compile (run without errors). It should be easy to follow and include the use of variables, conditions, loops, and functions where necessary. Comment to make the code easier to understand. Do not forget your name.

Rubric

70 Points - Project Report + Presentation

10 Points - Project Schematic

5 Points - Video Demo

15 Points - Project Code

You will be presenting your project presentations on September 29, 2026, so bring everything to class. You should expect to spend about 5-7 minutes with your presentation, including the project you chose, your solution, challenges faced along the way, and any important takeaways. Finally, you will demonstrate your project working in real time.

If you have any concerns with any part of the project, do not wait to reach out for clarification or help. You are welcome to use all resources available, including ChatGPT, but you must cite all sources of help. This is an individual project; however, if you receive help from your peers, that's great, but be sure to mention that in your report and/or sketch.